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# Quiet Secular Variation in Japan during the Last 9500 Years*

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## 1. Introduction

It now appears that secular variation of the geomagnetic field has been characteristically quiet in the central Pacific region during the last few million years (DOELL and COX, 1972; DOELL, 1972). From a very extensive paleomagnetic study of Hawaiian lava flows, they found that an angular standard dispersion of VGPs calculated for 219 lava flows of the Brunhes epoch is about 10.8°. This value is significantly smaller than a total angular standard dispersion calculated as the sum of an angular standard dispersion of 11.5° assigned to a dipole wobble and that for the present (1965 IGRF) non-dipole components at the same latitude. If the above-mentioned total angular standard dispersion represents the secular variation of the geomagnetic field during the last few million years, it is obvious that the secular variation in Hawaii is essentially due to dipole wobble and has almost no nondipole components. DOELL and COX (1972) also showed that secular variation in Galapagos, New Zealand, France, Alaska, Iceland and Antarctica during the Brunhes epoch is significantly higher than in the central Pacific, the angular standard dispersion calculated for the former regions lying approximately on the above-mentioned total angular standard dispersion curve (see Fig. 1).

COX and DOELL (1964), HIDE (1967), and DOELL and COX (1972) suggested that the temperature distribution in the lower mantle or the topography of the core-mantle interface beneath the central Pacific would act to suppress the nondipole field. Although there is no direct support for this hypothesis, it is obvious that the existence of such a characteristic quiet secular variation region gives an important constraint to the theory of the geomagnetic field. Hence, it is very desirable to examine whether

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such low secular variation is characteristic only of the central Pacific region or if it is to be found in other regions as well. In order to approach the problem, we have calculated the angular standard dispersion in Japan during the last 9500 years.

## 2. Secular Variation in Japan during the Brunhes Epoch

### *2.1 Secular variation in archeomagnetic data during the last 9500 years*

In order to obtain meaningful information about secular variation, paleomagnetic data should extend over more than a few thousand years. Spectral analyses of the geomagnetic field show that nondipole components are characterized by periods of up to a few thousand years (DOELL and COX, 1972). It is also well known that the main field and its secular change drift westward with a velocity of 0.2 to 0.4 degree year$^{-1}$, that is, with periods of 1800 to 900 years (YUKUTAKE, 1970). Absence of the non-dipole field or persistent low secular variation can be assumed only if there has been long-term absence of the angular standard dispersion for at least several thousand years. The most crucial part in establishing such long-term absence of non-dipole components is then in ascertaining the time span involved. Hence, in carrying out the present investigation, our chief concern was to choose paleomagnetic data for which the time involved was ascertained to be long enough ($>10^3$ years) to give meaningful information on secular variation.

There are 4757 individual archeomagnetic measurements from 169 sites in Japan, their ages ranging from the present to 9500 B.P. Most of the measurements were performed on baked earth and the ages were well established on the basis of archeological and/or $^{14}C$ dating. Hence, there is little doubt that the secular variation calculated for these data represents the characteristic long-term feature of the geomagnetic field. The experimental data for individual measurements are found in a recent review article by KINOSHITA (1970). The calculation of an angular standard dispersion was done following the procedures described by DOELL and COX (1972), that is, an angular standard dispersion of VGPs was calculated with respect to the present geographical axis designated as $S_T$ by DOELL and COX (1972). This procedure enables us to make a direct comparison of the present results with those ($S_T$) by DOELL and COX (1972). The VGPs determined for the 169 sites are divided into 28 groups, each having an equal interval of 100 years, and an average VGP for each interval was then calculated. The purpose of the latter procedure is to avoid unfair weight which might be assigned to a particular interval having an excessively large number of measurements. Table 1 shows archeomagnetic data for each group, (1) age, (2) number of measurements, (3) mean group VGP

Table 1. Archeomagnetic Data in Japan

| Age | N* | VGP | | $\alpha_{95}$° |
|---|---|---|---|---|
| | | lat. | long. | |
| A.D. 1900–Present | 1 | 82.2 | 16.6 | — |
| 1700–1799 | 2 | 78.0 | 10.0 | 8.5 |
| 1600–1699 | 2 | 72.6 | −80.6 | 2.7 |
| 1500–1599 | 3 | 74.1 | −71.2 | 6.2 |
| 1400–1499 | 2 | 81.9 | −25.1 | 10.2 |
| 1300–1399 | 10 | 82.6 | −83.0 | 4.3 |
| 1200–1299 | 8 | 83.5 | −96.0 | 4.3 |
| 1100–1199 | 7 | 83.5 | 0.0 | 2.8 |
| 1000–1099 | 7 | 80.4 | 25.3 | 8.4 |
| 900– 999 | 9 | 73.4 | 29.1 | 2.4 |
| 800– 899 | 14 | 75.2 | 20.0 | 2.3 |
| 700– 799 | 21 | 77.4 | 31.3 | 1.7 |
| 600– 699 | 14 | 75.3 | 25.0 | 2.4 |
| 500– 599 | 11 | 76.0 | 21.0 | 5.0 |
| 400– 499 | 16 | 81.6 | −15.0 | 2.9 |
| 300– 399 | 8 | 85.9 | −69.7 | 4.8 |
| 200– 299 | 5 | 83.6 | −89.7 | 3.8 |
| B.C. 2600–2501 | 2 | 84.5 | −92.5 | 3.3 |
| 2800–2701 | 5 | 86.1 | −70.2 | 7.4 |
| 3100–3001 | 3 | 87.1 | −77.5 | 11.2 |
| 3200–3101 | 2 | 86.6 | −57.3 | 11.4 |
| 3300–3201 | 3 | 85.1 | −41.4 | 2.6 |
| 3400–3301 | 2 | 84.2 | −31.5 | 11.4 |
| 3500–3401 | 1 | 77.2 | −13.3 | — |
| 3700–3601 | 3 | 78.8 | 22.1 | 10.8 |
| 4100–4001 | 6 | 80.8 | −90.2 | 6.0 |
| 7100–7001 | 1 | 84.2 | 31.9 | — |
| 7600–7501 | 1 | 74.8 | 15.9 | — |

* Number of sampling site

Table 2. Angular Standard Dispersion Calculated for Japanese Archeomagnetic and Paleomagnetic Data

| Data | Age | N | $\alpha_{95}$° | $S_T$ |
|---|---|---|---|---|
| Japanese archeomagnetic | | | | 3.6°[5] |
| Sample | 0–9500 B.P.[1] | 28[4] | 2.7° | $10.3°^{+2.3°}_{-1.6°}$ |
| Hakone OSN flows | Middle Brunhes epoch[2] | 23 | 5.9° | |
| Usami lava flows | 0.7–0.9 m.y.[3] | 11 | 8.6° | |

1) $^{14}C$ dating and archeological information (KINOSHITA, 1970)
2) Geological estimate (H. KUNO, private communication, 1962)
3) K-Ar dating (KANEOKA et al., 1971)
4) Number of intervals (see text)
5) Errors are 95% confidence limits

direction and (4) within group dispersion $\alpha_{95}$°. Finally, a total angular standard dispersion for the 28 averaged VGPs was calculated with respect to the present geographical axis. The angular standard dispersion thus calculated is 10.3°, which is shown in Table 2. In Fig. 1, the result is compared with the total angular standard dispersion curve and the results obtained by COX and DOELL (1972). It is remarkable that the secular variation obtained for the Japanese archeomagnetic data for the last 9500 years is as small as that for Hawaii in spite of the higher latitude of the former site.

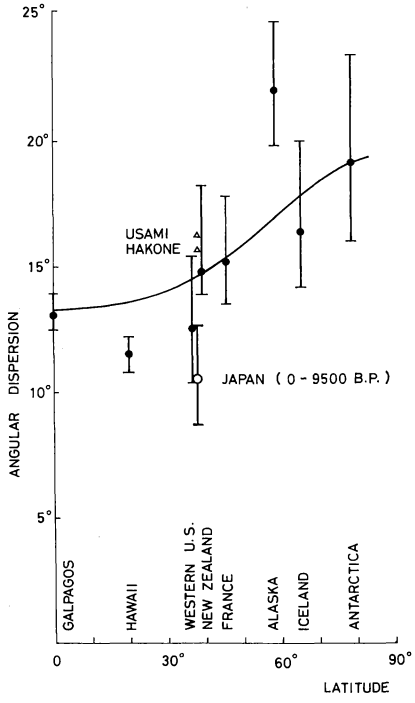



Fig. 1 Angular standard dispersion $S_T$ of virtual geomagnetic poles calculated with respect to the geographic axis. Solid curve: total angular dispersion (see text), error bars indicate 95% confidence limits. (Except the Japanese data, the figure is redrawn from DOELL and Cox (1972)). Angular standard dispersion $S_T$ for Usami and Hakone are shown by open triangles to indicate that the time span for the data may not be long enough to represent long-term secular variation.

### 2.2 *Secular variation on paleomagnetic data from Hakone and Usami lava flows*

In Fig. 1, paleomagnetic data obtained for Hakone and Usami lava flows of Brunhes epoch are also presented. K-Ar ages of the Usami flows range from 0.7 to 0.9 m.y. (KANEOKA et al., 1970). Although, there are no radiometric data for the Hakone lava flows, its age is estimated to be the middle Brunhes epoch on the basis of geological observation (KUNO, private communication, 1962). Kuno also estimated that in both cases the total time span for the whole group of flows should be at least several thousand years judging from weathered interbedding surfaces between the lave flows. Since there is no other evidence for this estimation, the secular variation obtained on these flows should be regarded as preliminary. It is, however, interesting that the values of angular standard dispersion for Usami and Hakone flows are significantly larger than that for the archeomagnetic data (Fig. 1), the former lying on the total angular standard dispersion curve proposed by DOELL and COX (1972).

It should be noted here that some of the archeomagnetic data used in the present calculation lack a-c demagnetization treatment and thus experimental precision for the archeomagnetic data may be inferior to that for the paleomagnetic measurements of Usami and Hakone lava flows. However, stability of most samples is ascertained by a tight within-site grouping of the NRM direction in which $\alpha_{95^\circ}$ is generally less than 10°, and is verified by storage test (see KINOSHITA, 1970). In addition secondary magnetization would in general act to disperse NRM directions resulting in an apparently larger secular variation. Hence, we may conclude that the angular standard dispersion calculated for the archeomagnetic data in fact represents the long-term quiet secular variation of the geomagnetic field in Japan during the last 9500 years.

## 3. Discussion

If a particular configuration of the core-mantle interface is responsible for the long-term absence of nondipole field as suggested by COX and DOELL (1964), HIDE (1967) and DOELL and COX (1972), the observed low secular variation in Japan requires that it should not be limited only to a region corresponding to the central Pacific.

Alternatively, it is possible that low secular variation is rather a general feature of the geomagnetic field in the last ten thousand years during which the Japanese archeomagnetic samples and the Kau lava flows (Hawaii) were formed. Then the low secular variation found in the central Pacific as well as in Japan would not indicate any space characteristics of the geomagnetic field, but rather represent its long-term time charac-

teristics. In this connection, it is particulary interesting to note the analysis for recent data by YUKUTAKE (private communication, 1972) suggests that the dipole wobble may be as small as a few degrees (say, 4–6°). The coincidence of the angular standard dispersion for Galapagos, New Zealand, France, Iceland and Antarctic paleomagnetic data with the total angular standard dispersion curve calculated for a dipole wobble of 11.5° then may indicate that the dipole wobble was in fact as large as 11.5° during the period when these lava flows were formed, that is, during the early to middle Brunhes epoch. If this interpretation is correct, large secular variation, if found in the last thousand years, should reflect either magnetic instability of rocks or local magnetic disturbances.

If we take a second view that the low secular variation is a manifestation of time characteristics of the geomagnetic field rather than space characteristics, it may be possible to suppose, as a third possibility, that a larger secular variation in the early Brunhes epoch than in the last several thousand years reflects integrated tectonic movements in a given area, since such an effect should be primarily a function of time and results in larger dispersion of the NRM directions with increasing age. Recently DOELL (1972) noted that paleosecular variation increases with age of the lavas in Hawaii, France and Iceland. It is then important to note that such a relation with age is found irrespective of the absolute age of the lavas. These observation then may be most easily explained in terms of an effect of integrated tectonic movements. However, this interpretation does not explain almost the same angular standard dispersion in the Matuyama to the early Brunhes epoch as in the last ten thousand years in Hawaii (DOELL, 1972), unless one assumes persistently quiet tectonic disturbance in this area.

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